Question 1

1NF

order											
customerName	**customerNum (PK)**	customerType	**date (PK)**	**time (PK)**	employee	**partNum (PK)**	name	type	cageCode	quantityOrdered	unitPrice
Jeff Peterson	HG5487	Consumer	7/1/2024	10:30 AM	D. Harrison	10654	Float Control	Plumbing	G413	4	12
Jeff Peterson	HG5487	Consumer	7/1/2024	10:30 AM	D. Harrison	10456	Modulator	Electrical	H433	3	7
Jeff Peterson	HG5487	Consumer	7/1/2024	10:30 AM	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
Jeff Peterson	HG5487	Consumer	7/1/2024	10:30 AM	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

2NF

customer		
customerNum (PK)	customerName	customerType
HG5487	Jeff Peterson	Consumer

order			
customerNum (PK,FK)	**date (PK)**	**time (PK)**	employee
HG5487	7/1/2024	10:30 AM	D. Harrison

part				
partNum (PK)	name	type	cageCode	unit_price
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10766	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

part_ordered				
customerNum (PK, FK)	**date (PK)**	**time (PK)**	***partNum (PK, FK)***	quantityOrdered
HG5487	7/1/2024	10:30 AM	10654	4
HG5487	7/1/2024	10:30 AM	10456	3
HG5487	7/1/2024	10:30 AM	10776	7
HG5487	7/1/2024	10:30 AM	106576	5

3NF

customer		
customerNum (PK)	customerName	customerType
HG5487	Jeff Peterson	Consumer

order			
customerNum (PK,FK)	**date (PK)**	**time (PK)**	employee
HG5487	7/1/2024	10:30 AM	D. Harrison

part			
partNum (PK)	name	*type (FK)*	unit_price
10654	Float Control	Plumbing	12
10456	Modulator	Electrical	7
10766	Hose Assembly	Plumbing	9
10657	Float Assembly	Plumbing	10

type_location	
type (PK)	cageCode
Plumbing	G413
Electrical	H433

part_ordered				
customerNum (PK, FK)	**date (PK)**	**time (PK)**	***partNum (PK, FK)***	quantityOrdered
HG5487	7/1/2024	10:30 AM	10654	4
HG5487	7/1/2024	10:30 AM	10456	3
HG5487	7/1/2024	10:30 AM	10776	7
HG5487	7/1/2024	10:30 AM	106576	5

A)

- The same person is ordering all the parts
- The parts are ordered at the same time
- The customer type will be the same for each item ordered and for the subsequent orders

B) The table starts out in 1NF because all the values are atomic and there are no repeating groups. The primary key is a composite key made up of customerNum, date, time, and partNum since the combination of these four attributes make each unique. To get the database into 2NF, I had to split up the parital dependencies. The first partial dependency is customerName and customerType are only dependent on customerNum. The second partial dependency is name, type, cageCode, and unitPrice only depend on the partNum. The third partial dependency is quantityOrdered is dependent on the complete composite key. To get the database into 3NF, transitive dependencies have to removed. The transitive dependency in the database is that cageCode is dependent on type, so type and cageCode had to be moved to a separate table.

Question 2

Unnormalized

pancea_corp					
staffNo (PK)	therapistName	**patNo (PK)**	patName	**appointmentDateTime (PK)**	branchNo
S1011	Fred SMith	P100	Lily Whihte	9/12/2022, 10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022, 12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022, 10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022, 14:00	Q10
S1032	Richrd Levin	P105	Jill Baker	9/14/2022, 16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022, 18:00	B13

1NF

pancea_corp						
staffNo (PK)	therapistName	**patNo (PK)**	patName	**appointmentDate (PK)**	**appointmentTime (PK)**	branchNo
S1011	Fred Smith	P100	Lily Whihte	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richrd Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

2NF

therapist	
staffNo (PK)	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

patient		
patNo (PK)	patName	branchNo
P100	Lily White	M15
P105	Jill Baker	M15
P108	Andy McKee	Q10
P110	Jimmy Winter	B13

appointment			
staffNo (PK,FK)	***patNo (PK,FK)***	**appointmentDate (PK)**	**appointmentTime (PK)**
S1011	P100	9/12/2022	10:00
S1011	P105	9/12/2022	12:00
S1024	P108	9/12/2022	10:00
S1024	P108	9/14/2022	14:00
S1032	P105	9/14/2022	16:30
S1032	P110	9/15/2022	18:00

3NF

therapist	
staffNo (PK)	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

patient_id	
patNo (PK,FK)	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

patient_branch	
patNo (PK,FK)	branchNo
P100	M15
P105	M15
P108	Q10
P110	B13

appointment			
staffNo (PK,FK)	***patNo (PK,FK)***	**appointmentDate (PK)**	**appointmentTime (PK)**
S1011	P100	9/12/2022	10:00
S1011	P105	9/12/2022	12:00
S1024	P108	9/12/2022	10:00
S1024	P108	9/14/2022	14:00
S1032	P105	9/14/2022	16:30
S1032	P110	9/15/2022	18:00

Assumptions:

- Patients can only go to one branch
- Patients can only have one appointment a day
- Therapists can only see one patient at a time
- Therapists can work at multiple branches

B) The table is unnormalized because the appointmentDateTime is not atomic. However, it is a primary key along with staffNo and patNo since it makes each row unique. To achieve 1NF, appointmentDateTime needs to split up into appointmentDate and appointmentTime. To achieve 2NF, partial dependencies need to be split up. The partial dependencies are 1) therapistName is only dependent on staffNo; 2) patName and branchNo are only dependent on patNo 3) staffNo, partNo, appointmentdate, and appointmentTime need to be in their own table because the whole composite key doesn't determine any other attribute but still needs to be recorded. To achieve 3NF, transitive dependencies need to be removed. Since branchNo is dependent on patName (a non-key attribute) and patNo, the patient table has to split up into a patient_id table and patient_branch table. By splitting these tables up, this reduces anomalies. For example, in the patient table, closing the B13 branch would remove Jimmy Winter from the database. Similarly, if Jimmy Winter left the practice, it would remove the B13 branch.

Question 3

1NF

maid_better					
eNo (PK)	**contractNo (PK)**	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	15	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

2NF

employee	
eNo (PK)	eName
1135	Smith J
1057	Hocine D
1068	White T

event	
eventNo (PK)	eventLoc
H25	Queens
H4	Yonkers

contract	
contractNo (PK)	*eventNo (FK)*
C1024	H25
C1025	H4
C1026	H25

employee_working		
eNo (PK, FK)	***contractNo (PK, FK)***	hours
1135	C1024	16
1057	C1024	24
1068	1025	15
1135	C1025	15
1135	C1026	10

3NF

employee	
eNo (PK)	eName
1135	Smith J
1057	Hocine D
1068	White T

event	
eventNo (PK)	eventLoc
H25	Queens
H4	Yonkers

contract	
contractNo (PK)	*eventNo (FK)*
C1024	H25
C1025	H4
C1026	H25

employee_working		
eNo (PK, FK)	***contractNo (PK, FK)***	hours
1135	C1024	16
1057	C1024	24
1068	1025	15
1135	C1025	15
1135	C1026	10

Assumptions

- Employees can work the same event under multiple contracts
- Employees can work different hours for different contracts
- Multiple employees can work the same event
- Contracts can be deleted
- Employees can be removed from contracts
- Employee hours can change
- Event locations can change

B) The table is in 1NF because there are no repeating groups and each value is atomic. The primary keys are eNo and contractNo because they make each row unique. To achieve 2NF, the partial dependencies need to be removed. The partial dependency is that eName is only dependent on eNo, so there needs to be a separate table for that. In achieving 2NF, eventNo and eventLoc need to have their own table because eventLoc is not dependent on eNo and contractNo; it is dependent only on eventNo. In doing this, deleting a contract (like C1025) doesn't delete an event (like H4). There also needs to be a separate table for eNo, contractNo, and hours since hours are only dependent on eNo and contractNo. However, in removing partial dependencies and creating separate tables, transitive dependencies are removed as well. So, in achieving 2NF, 3NF is achieved as well.